

SWE 303 – Static Testing Report

Student: Mikael Xhangolli

Project: Electronic Store (JavaFX)

Tools Used: SpotBugs, SonarQube for IntelliJ IDEA

1. Overview of Static Testing

Static testing was performed to identify potential code quality, maintainability, and design issues without executing the application. SpotBugs and SonarQube were used to analyze the source code. The goal was to improve code quality where appropriate and document justified warnings where changes were not suitable, in accordance with SWE 303 project guidelines.

2. Static Testing Activities Performed

- Ran SpotBugs analysis to detect potential bugs and bad practices.
- Ran SonarQube analysis to detect code smells and maintainability issues.
- Fixed issues that could be resolved without affecting system design.
- Justified remaining warnings related to UI design decisions.

3. Justification of Remaining SonarQube Issues

Issue 1: Extract nested code block into a method

The nested code is part of JavaFX event handling logic. Keeping the logic within the same method improves readability and preserves the flow of UI-related operations. Extracting the code into multiple methods would increase complexity and reduce clarity without providing functional benefit.

Issue 2: Remove unused method parameters (error, s, t, location)

These parameters are intentionally kept to maintain compatibility with existing method signatures and to support potential future extensions of the UI logic. Removing them could require unnecessary refactoring later. The parameters do not affect runtime behavior or performance.

Issue 3: Remove unused method parameter 'query'

The 'query' parameter is reserved for future search or filtering functionality within the dashboard. Maintaining the parameter ensures interface consistency and avoids breaking changes if the feature is implemented later.

Issue 4: Change instance reference to static reference

An instance reference is intentionally used to preserve object-oriented design principles. Using static references in the UI layer would increase coupling and reduce flexibility, making future maintenance and testing more difficult.

4. Summary of Work Done

All static testing tasks assigned were completed. SpotBugs and SonarQube were successfully executed, identified issues were reviewed, fixable issues were resolved, and remaining warnings were properly justified and documented. The static testing process improved code clarity and ensured compliance with SWE 303 project requirements.